

Ethan Mathews

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EDUCATION

Bren School of Environmental Science & Management

Santa Barbara, CA

Master of Environmental Data Science

Cumulative GPA: TBD

August 2026 - June 2027

University of California, Santa Barbara (UCSB)

Santa Barbara, CA

Bachelor of Arts (B.A.), Environmental Studies

Cumulative GPA: 3.98

September 2023 - August 2025

Mira Costa Community College (MC)

Cardiff, CA

Associate of Arts, Math and Science

Cumulative GPA: 3.94

August 2020 - June 2023

Relevant Coursework

Environmental Impact Analysis (A & B)

- Analyzed hypothetical development project components throughout Southern and Central California using published environmental data and personal field observations.
- Wrote a focused environmental impact report featuring biological resources that was a part of a mixed use faculty and classroom project located in an environmentally sensitive open space.
- Developed project descriptions, environmental settings, impact assessments, mitigation measures, and reporting plans for biological resources, in accordance with CEQA, to ensure comprehensive and defensible environmental review.
- Gained extensive public presentation skills by crafting and responding to peer Draft EIR, comment letters, addressing environmental stakeholder concerns.

Data Science in Environmental Studies

- Conducted statistical analyses and data visualizations using R and RStudio to explore environmental datasets of up to 200,000 observations, enhancing data-driven decision-making skills and understanding of environmental research methodologies.
- Developed reproducible workflows and collaborated on code repositories through GitHub, adhering to a rigorous approach to data science and improving collaborative coding practices.
- Applied statistical techniques, including hypothesis testing and regression analysis, to investigate environmental issues, strengthening the ability to draw meaningful conclusions from complex data.
- Created an HTML website as a professional academic portfolio to showcase past projects, academic progress, and future plans, utilizing Quarto and Markdown documents to enhance technical and presentation skills.

Geographic Information System (A, B, and C)

- Led a team to develop and execute comprehensive GIS projects analyzing income-based access to green space, conducting land-suitability modeling for a new fire station in Paradise, California, and assessing nitrate contamination in domestic wells in proximity to almond and grape orchards in Fresno County, demonstrating strong leadership and collaboration through successful presentations to professors and peers.
- Applied GIS skills to research questions and spatial analysis tasks to increase proficiency in handling complex geographic problems, improving decision-making based on statistical analysis.
- Analyzed digital geographic data using algorithms and analytical procedures to enhance spatial analysis accuracy and data interpretation.

- Investigated technical GIS issues including modeling approaches and fitting models, to optimize problem-solving in spatial data analysis and improve data handling efficiency.

Environmental Law

- Developed a foundational understanding of constitutional principles and major federal environmental laws; including the Clean Water Act, Clean Air Act, National Environmental Policy Act, Endangered Species Act, and Safe Drinking Water Act, along with their ethical, cultural, and legal foundations, strengthening my comprehension of environmental regulation.
- Analyzed administrative and judicial processes in environmental policy and litigation, strengthening understanding of agency roles in implementing statutes.
- Facilitated meaningful discussions with guest lecturers from various environmental agencies in the Santa Barbara area, enhancing understanding of coordination with state and federal agencies.
- Enhanced legal analysis and critical thinking skills by evaluating environmental law case studies and assessing their impact on regulatory compliance and agency decision-making.

ACADEMIC EXPERIENCE

Research Assistant to PhD Candidate Kelly Garvey January 2025-Present

- Conducted outreach to over 1,500 well drillers, collecting valuable data on drilling practices, pricing, and materials used, contributing to research on how drilling costs vary by county due to differing lithologies.
- Developed and applied Python coding skills using Google Collab notebooks to merge, clean, and visualize large datasets (200,000–1,000,000 observations) from the California Open Data Portal, as part of creating exploratory data as well as an interactive database.
- Scraped and compiled data from federal and state agency websites to identify and characterize injection wells across the United States, contributing to a broader analysis of permitting trends and geospatial patterns.
- Developed proficiency in data organization and project management, streamlining outreach workflows and enhancing data collection strategies for effective engagement with key stakeholders.

Research Assistant to Dr. Debra Perrone and Associate Professor Rebecca Nelson November 2025-Present

- Conducted a comprehensive review of Kansas groundwater administrative policies (1980s–1990s) to identify the evolution of regulatory controls affecting groundwater availability and basin management.
- Analyzed primary-source administrative procedures and statutory citations to determine relevance, scope, and legal implications for groundwater permitting.
- Built a structured, multi-variable database tracking policy iterations, amendments, supersessions, and basin-specific effects across 25+ administrative procedures.

WORK EXPERIENCE

Manager, Encinitas Fish Shop, Encinitas June 2021-Present

- Managed 20 employees over 5-8 hour work shifts in a 36-table, exclusive culinary setting having been promoted to a management position within the first year of employment.
 - Prioritizing customer satisfaction by managing a dynamic and high-pressure setting and achieving daily sales exceeding \$25,000.
 - Navigating challenging customer interactions and restaurant incidents, sharpening expertise in conflict resolution and the diplomatic management of difficult situations.
 - Collaborating with management team to improve team efficiency by reinforcing procedural compliance and optimizing staff performance, resulting in a reduction of labor costs from 25% to 10.5%-12.5%.
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EXTRACURRICULAR INVOLVEMENT

Member, of Environmental Agency Board (EAB)

- United as a group to assess current environmental incidents and discuss how the organization can support various causes.
- Organized local outings that ignite community engagement and conveyed the organization's objectives while increasing its impact on nearby communities including involvement in the community pushback of the proposed Sable Pipeline and biweekly beach cleanups.
- Furthered the group's local ecological knowledge that increased helps the adjacent communities understanding of ecological interactions that have benefitting natural resource use practices.

Member, UCSB Environmental Law Club (ELC)

Member, National Ground Water Association (NGWA)

Intramural Sports (Basketball & Soccer)

SKILLS

Languages: Spanish (Intermediate)

Software: Proficient with Geographic Information Systems, Python, and R.

Technical Lab Techniques: Data analysis, technical writing, field sampling (GrayWolf particulate meter, Oakton Probe, Ion-Selective Electrode, and Colorimeter).